

# The Law of Laws: Statistical Lawhood Under Finite Description

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(Dated: May 13, 2026)

Physics is written at scales that have already discarded information. This paper asks which statistical form of law can remain stable under that discarding. In a finite positive-support setting, four requirements—relabelling invariance, locality of statistical content to individual outcomes (*atomwise locality*), independence of arbitrary cell-size choices (*neutral refinement*), and staged elimination—force scalar law-content to be proportional to relative entropy. Adding visible constraints then gives the free-energy variational family, with ordinary renormalisation appearing as generator flow inside that fixed family rather than as the origin of the family itself. A finite theory implemented inside the world it describes also yields non-unique reconstruction maps (the micro-to-macro lifting is not canonical). The failure of exact descent,  $\mathcal{O} = CG_\mu - G_MC$ —the commutator deficit between evolving and then coarse-graining versus coarse-graining and then evolving—composes across scales and provides a common algebraic language for residual couplings, memory kernels, gauge-like reconstruction freedom, holonomy, and incompatible finite descriptions. The result is not a derivation of quantum mechanics or the Born rule. It is a finite theorem and calculus for what an embedded statistical theory must preserve, lose, and record for its losses to remain law-governed.

## I. INTRODUCTION

Successful physical theories are rarely written in microscopic variables alone. Hydrodynamics, statistical mechanics, open-system dynamics, effective field theory, and renormalisation all replace a high-resolution description by a smaller one that still closes well enough to predict. In operator language, the basic demand is that a coarse map  $C$ , a fine generator  $G_\mu$ , and an effective generator  $G_M$  nearly form a commuting square,

$$CG_\mu \approx G_MC. \quad (1)$$

The approximation is the interesting part. Eliminated variables do not disappear; they return as effective couplings, memory terms, noise, curvature, or contextual dependence.

This paper asks a slightly prior question. What form must a statistical law have if it is to remain a law after hidden refinements are eliminated? The answer developed below has two parts. First, in the finite positive-support setting, the scalar content that can be decomposed through staged refinements is relative entropy. With visible constraints, this gives the usual free-energy/partition-function form. Second, when the theory is a finite register inside the world it describes, coarse-graining creates *reconstruction fibres*—for each macrostate, the set of microstates mapped to it under the coarse description. The failure of exact descent then becomes a composable *obstruction* (the commutator deficit  $\mathcal{O} = CG_\mu - G_MC$ ) rather than an unstructured error; and it accumulates across scale towers in a law-like way.

These are familiar phenomena seen from a common angle. Renormalisation replaces eliminated degrees of freedom by coupling flow, fixed points, and irrelevant directions [1–6]. Effective field theory organizes the same lesson through symmetry-allowed operator bases and controlled truncations [7, 8]. Projection methods and influence functionals show that eliminated environment variables reappear as memory kernels and influence phases

[9–13]. The common structure is not loss alone, but loss whose residue remains representable.

The variational side of the story is equally old. Entropy maximisation and free-energy minimisation give equilibrium statistical mechanics [14, 15]; maximum-caliber and large-deviation formulations extend the same variational grammar to paths and fluctuations [16, 17]; thermodynamic formalism expresses related equilibrium-state and pressure principles in dynamical language [18]. The present contribution is not another derivation of equilibrium thermodynamics. It identifies the finite compositional requirement that selects the relative-entropy/free-energy law-form before a particular physical Hamiltonian or dynamics is specified.

The reflexive case makes the need for such a formulation sharp. Let  $\Omega$  denote the configuration space of the world, let  $R \subset \Omega$  be the physical register carrying a theory, and let  $\pi : \Omega \rightarrow X$  be the symbol map by which that register represents the world. A detached laboratory model may treat its register as outside the target. A universal or cosmological model cannot: its target is  $\Omega$  itself, so

$$R \subset \Omega, \quad Y = \Omega, \quad \pi : \Omega \rightarrow X, \quad |X| < |\Omega| \quad (2)$$

for any nontrivial compression. Exact reflexive description would require a finite compressed image to contain its own representational action without residue. Generically, it cannot. The residue is not a defect to be hidden; it is the object whose composition laws determine whether a finite theory can remain predictive.

Geometry and quantum foundations provide the cautionary examples for the second half of the paper. Berry phase, holonomy, and non-Abelian geometric phase show that locally unobservable section choices can leave loop-level traces [19–21]. Information geometry and the quantum geometric tensor show the complementary split between symmetric distinguishability and antisymmetric curvature [22–24]. Meanwhile, quantum logic, Bell, Kochen-Specker, and sheaf-theoretic contextuality warn

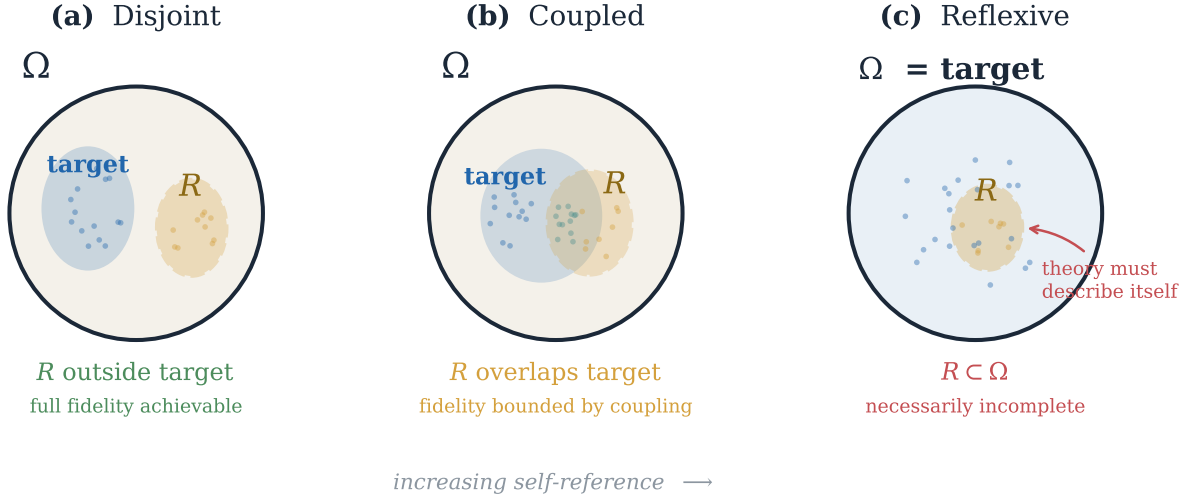


FIG. 1. Representational regimes. In the reflexive case the register  $R$  is a proper subsystem of the world it represents,  $R \subset \Omega$ , while the target is  $\Omega$  itself. A finite reflexive theory is exact only when the compressed self-image closes self-similarly. Otherwise the residual is structured loss, represented below by descent obstructions.

that one cannot generally flatten all contexts into a single Boolean catalogue of primitive values [25–29].

The claim below is therefore bounded but concrete. The paper derives a finite relative-entropy/free-energy law-form from compositional elimination, then develops a finite obstruction calculus—a systematic accounting of commutator deficits  $\mathcal{O} = CG_\mu - G_MC$  across scale towers—for embedded descriptions. It does not derive Hilbert space, the Born rule, or a primitive ontology. Those would require additional assumptions, and the point of the v2 presentation is to say exactly where they would enter.

## II. CLAIMS AND ROUTE

The argument is organized around two invariance requirements. The first is statistical: eliminating hidden refinements should not change the kind of scalar law-content being used. The second is representational: when exact descent fails, the failure must compose coherently across scales. Table I states the resulting claims and their status. Readers unfamiliar with the terminology (*obstruction*, *atom*, *refinement fibre*, etc.) may consult Table II for the standard statistical-mechanics equivalents before reading further.

The first chain of reasoning is

$$\begin{aligned}
 &\text{stable statistical law under refinement} \\
 &\implies \text{recursive scalar law-content} \\
 &\implies \text{KL/entropy} \implies \text{free energy.}
 \end{aligned} \tag{3}$$

The key assumption is staged elimination: hidden refinements may be removed in one step or in several steps without changing the kind of content assigned. Relabelling and locality first reduce admissible finite content

to a probability-weighted function of local likelihood ratios. The staged chain rule then forces the logarithm. Once visible constraints are included, the Legendre-dual variables assemble into a generator  $K$ , and the stable family becomes

$$J_K[\rho] = \text{Tr}(\rho K) - MS(\rho). \tag{4}$$

Ordinary RG then acts as  $K \mapsto K_{\text{eff}}$  inside this family.

The second chain of reasoning is

$$\begin{aligned}
 &\text{finite reflexive description} \\
 &\implies \text{coarse fibres and reconstruction freedom} \\
 &\implies \text{descent obstruction} \\
 &\implies \text{geometry of residual freedom.}
 \end{aligned} \tag{5}$$

The central object is the failure of the fine and coarse dynamical routes to agree,

$$\mathcal{O} = CG_\mu - G_MC. \tag{6}$$

When  $\mathcal{O} = 0$ , the macro-description is an autonomous quotient. When  $\mathcal{O} \neq 0$  but composes coherently, its visible part can appear as renormalisation, while its unresolved part can appear as memory, induced interactions, curvature, or incompatible event structure.

The paper should be read with this scope in view. It proposes a finite statistical and algebraic spine for law-hood under coarse description. It does not ask the reader to accept quantum mechanics as already derived from that spine. Instead, it identifies which additional geometric and probabilistic assumptions would be needed to make such a reconstruction program precise.

TABLE I. Logical status of the main results. The finite theorems are proved in the text and Appendix A; the quantum and geometric statements are bounded consequences or comparison principles, not reconstructions of quantum mechanics.

| Claim                            | Status                             | Role                                      |
|----------------------------------|------------------------------------|-------------------------------------------|
| Local ratio form                 | finite proposition                 | relabelling, locality, neutral refinement |
| KL law-content                   | finite proposition                 | staged refinement chain rule              |
| Free-energy family               | finite proposition                 | visible constraints plus KL content       |
| RG flow                          | finite contraction identity        | generator flow inside invariant law-form  |
| Reconstruction freedom           | finite affine-geometry proposition | sections of a coarse map                  |
| Obstruction composition          | finite algebraic identity          | law-like residuals across scales          |
| Memory kernels                   | exact finite/block identity        | hidden variables returning in time        |
| Gauge/holonomy language          | conditional geometry               | requires a connection on reconstructions  |
| Noncommuting erasures            | finite example                     | route to non-Boolean event algebra        |
| Born rule/Hilbert reconstruction | not derived                        | requires additional assumptions           |

### III. STATISTICAL LAWS AS VARIATIONAL OBJECTS

Let  $r \in \mathcal{R}$  denote a representation or descriptive resolution, and let  $\rho_r$  be a statistical state expressible at that resolution. We define a statistical law at  $r$  as a variational selector.

**Definition 1** (Law-functional). *A statistical law at representation  $r$  is a functional  $J_r[\rho_r]$  whose minimisation selects the regular state*

$$\rho_r^* = \arg \min_{\rho_r} J_r[\rho_r]. \quad (7)$$

This definition places the law in the functional  $J_r$ , not only in the selected state  $\rho_r^*$ . Two functionals can share a minimiser while assigning different stability, response, fluctuation, and coarse-graining behaviour. The object that can survive a change of description is therefore the functional form.

Now let  $\pi : X \rightarrow Y$  be a coarse-graining from a fine state space  $X$  to a coarse state space  $Y$ . A fine state induces a coarse state  $\rho_Y = \pi_* \rho_X$ . For a fixed  $\rho_Y$ , the compatible fine states form a refinement fibre,

$$\{\rho_X : \pi_* \rho_X = \rho_Y\}. \quad (8)$$

The first natural induced coarse functional is the contraction over hidden refinements,

$$J_Y[\rho_Y] = \inf_{\rho_X : \pi_* \rho_X = \rho_Y} J_X[\rho_X] + \text{const.} \quad (9)$$

This is the variational analogue of integrating out variables.

Equation (9), by itself, is too weak. Any functional can be contracted. Lawhood requires the contracted object to remain in the same family of law-forms. If  $\mathfrak{F}$  is the family of admissible law-functionals, the closure condition is

$$\mathcal{C}_\pi(\mathfrak{F}) \subseteq \mathfrak{F}, \quad (10)$$

where  $\mathcal{C}_\pi$  denotes contraction over fibres of  $\pi$ . This is not an ordinary RG fixed point. It is an invariant functional family in which ordinary RG flows may later live.

**Assumption 1** (Self-similar elimination). *For a tower  $X \xrightarrow{\pi} Y \xrightarrow{\sigma} Z$ , eliminating hidden variables in stages must agree with eliminating them directly, up to the allowed reparameterisations of the law-family.*

This is the substantive requirement. Without it, a coarse-grained law is only what remains after erasure. With it, the erased part carries the same kind of statistical content as the visible part.

The next question is therefore not which distribution is most plausible at a single level. It is which scalar content can be split into visible and hidden conditional pieces without changing its kind. That is the chain-rule problem.

### IV. CHAIN RULE, ENTROPY, AND FREE ENERGY

Consider finite probability distributions  $p_X$  and  $q_X$  on  $X$ , and a coarse map  $\pi : X \rightarrow Y$ . The induced marginals are

$$p_Y(y) = \sum_{x:\pi(x)=y} p_X(x), \quad q_Y(y) = \sum_{x:\pi(x)=y} q_X(x). \quad (11)$$

Inside the fibre  $X_y = \pi^{-1}(y)$ —the set of all fine states mapped to the same macrostate  $y$ —define the conditionals  $p_{X|Y}(\cdot|y)$  and  $q_{X|Y}(\cdot|y)$ . Self-similar law-content is the requirement that a scalar  $\mathcal{I}(p||q)$  decompose as

$$\begin{aligned} \mathcal{I}_X(p_X||q_X) &= \mathcal{I}_Y(p_Y||q_Y) \\ &\quad + \mathbb{E}_{p_Y} \mathcal{I}_{X_y}(p_{X|Y}(\cdot|y)||q_{X|Y}(\cdot|y)). \end{aligned} \quad (12)$$

In words: statistical distinguishability between two theories must split correctly when the state space is partitioned—the total difference equals the coarse-level

TABLE II. Terminology glossary. The left column lists terms used in this paper; the right column gives the standard statistical-mechanics or operator-algebra equivalent. All terms on the left are formally defined in the text; this table is a reading aid, not a substitute for those definitions.

| Term used in this paper                          | Standard statistical-mechanics equivalent                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Atom</i> (Prop. 1)                            | Elementary outcome or microstate; a single sample point of the probability space $X$ .                                                                                                                                                                                                                                                                                                                     |
| <i>Refinement fibre</i> $\pi^{-1}(y)$            | Set of all microstates consistent with macrostate $y$ ; Boltzmann’s <i>complexion</i> class. Its log-cardinality under a uniform reference gives Boltzmann entropy $S = k_B \ln W$ .                                                                                                                                                                                                                       |
| <i>Coarse map</i> $C : X_\mu \rightarrow X_M$    | Marginalisation or projection operator; the operation of summing (tracing) over hidden variables in a partition function or density matrix.                                                                                                                                                                                                                                                                |
| <i>Reconstruction map</i> $R$                    | A right inverse of $C$ ; a choice of “lifting” from macro to micro. In practice, any conditional distribution $p(x m)$ that is consistent with the observed macrostate serves as a reconstruction.                                                                                                                                                                                                         |
| <i>Reconstruction freedom</i> $R' = R+N, CN = 0$ | Gauge freedom in the micro-to-macro lifting; the arbitrariness in choosing which microstates represent a given macrostate. Different choices correspond to different sections of the coarse-graining fibre bundle.                                                                                                                                                                                         |
| <i>Obstruction</i> $\mathcal{O} = CG_\mu - G_MC$ | Commutator deficit: the amount by which “evolve then coarse-grain” disagrees with “coarse-grain then evolve”. Depending on context it surfaces as a <i>memory kernel</i> (Mori–Zwanzig), a <i>beta-function correction</i> (renormalisation group), an <i>environment-induced coupling</i> (open-system dynamics), <i>geometric phase</i> (Berry holonomy), or <i>contextuality</i> (quantum measurement). |
| <i>Self-similar law-content</i>                  | The chain-rule property of KL divergence: $D_{\text{KL}}(p_X \  q_X) = D_{\text{KL}}(p_Y \  q_Y) + \mathbb{E}_{p_Y}[D_{\text{KL}}(p_{X Y} \  q_{X Y})]$ . This is the consistency condition that makes relative entropy the unique measure of statistical distinguishability that can be split between visible and hidden variables without remainder.                                                     |
| <i>Obstruction composition (1-cocycle)</i>       | The identity $\mathcal{O}_{02} = C_{12}\mathcal{O}_{01} + \mathcal{O}_{12}C_{01}$ for commutator deficits across a two-step scale tower. Obstruction accumulates like gauge curvature, not like a scalar error; the cross term is the signature of a 1-cocycle rather than a simple sum.                                                                                                                   |
| <i>Rank ensemble</i> $P_I(n) \propto n^{-I}$     | The maximum-entropy distribution subject to a fixed expected logarithmic cost $\langle \log n \rangle$ ; the Zipf or power-law ensemble. Its partition function $Z(I) = \zeta(I)$ is the Riemann zeta function.                                                                                                                                                                                            |
| <i>Law-functional</i> $J_r[\rho]$                | The variational free energy: the functional whose minimisation selects the physical state, generalising $F = \langle E \rangle - TS$ to arbitrary observable constraints and reference measures.                                                                                                                                                                                                           |

difference plus the average fine-level difference within each macrostate. This is the defining chain rule for KL divergence, and also the condition that makes KL the unique choice among all scalar law-contents.

Figures ?? and ?? make the geometry of this requirement unavoidable. The numerical evidence package verifies this identity for KL divergence to residual  $-2.78 \times 10^{-17}$ , while a simple  $L^2$  contrast on the same coarse map leaves residual  $-8.98 \times 10^{-2}$ . The example is not a uniqueness proof. It is a useful warning that recursive decomposition is a real condition.

The finite uniqueness result used here has one locality step before the chain rule is applied. Relabelling invariance and *atomwise locality* (the requirement that disjoint elementary outcomes contribute independently, each determined only by its own probability ratio  $p_i/q_i$ ) first

say that disjoint alternatives contribute by a common local density. A *neutral-refinement* clause then removes dependence on the arbitrary size of the cell: splitting an alternative into sub-alternatives with the same conditional weights under  $p$  and  $q$  adds no new law-content. Appendix A shows that these finite locality requirements force the local ratio form.

**Proposition 1** (Finite local law-content). *For finite positive distributions  $p, q$ , admissible scalar law-content has the form*

$$\mathcal{I}(p||q) = \sum_i p_i \varphi\left(\frac{p_i}{q_i}\right), \quad (13)$$

with  $\varphi$  continuous.

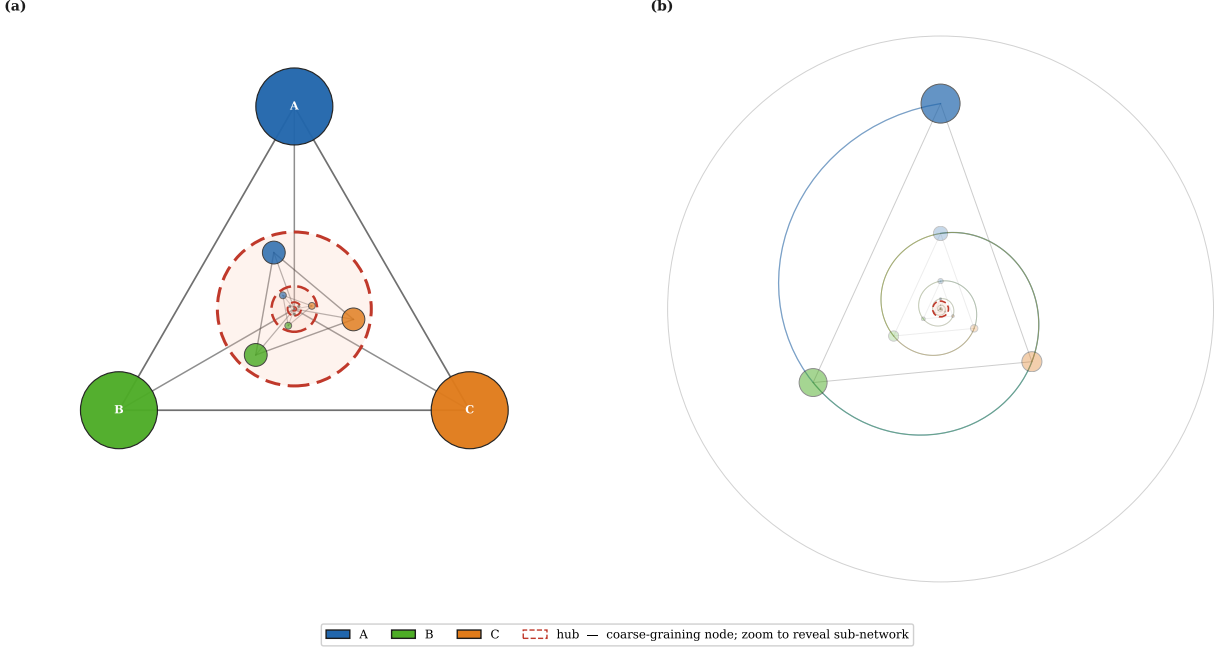


FIG. 2. **Recursive self-similarity as the forcing mechanism for the log kernel.** (a) A schematic inspired by Escher’s *Print Gallery*: each node in the network is itself a coarse-graining step with free-energy cost  $K_C = -M \log Z_C$ . Self-reference is not a paradox here; it is the structural demand that the cost of describing a coarse-graining step must be assignable by the same rule that assigns costs at every other level. This is the self-similarity condition applied to the law-functional: zooming in on any coarse-graining level must reproduce the same functional form. (b) Conformal log-spiral view: as costs accumulate inward toward a fixed point, the only kernel consistent with additive accumulation across a multiplicative rescaling of likelihood ratios is the logarithm. Any other continuous kernel fails the Cauchy functional equation  $\varphi(ab) = \varphi(a) + \varphi(b)$  (see Proposition 9).

This is the same kind of move used in the Cost of Complexity argument: relabelling removes arbitrary coordinates, while composition or neutral refinement fixes the scalar dependence that survives. Here the surviving local invariant is the likelihood ratio  $p_i/q_i$ , weighted by the amount of retained probability mass  $p_i$ .

**Proposition 2** (KL as recursively decomposable content). *For finite probability spaces satisfying the locality principle of Proposition 1, the chain-rule condition (12) fixes the scalar content up to scale as*

$$\mathcal{I}(p||q) = C \sum_i p_i \log \frac{p_i}{q_i}. \quad (14)$$

The finite local form and the logarithmic uniqueness step are proved in Appendix A. The theorem is therefore a theorem inside a declared finite locality principle: relabelling plus neutral refinement give the ratio-separable local density, and the staged-refinement chain rule then forces the logarithm. The resulting form is close to the information-type divergence setting studied by Csiszar [30, 31]. It is also aligned with older and newer characterisations of relative information, including prior-to-posterior uniqueness theorems, Shore–Johnson consistency, and functorial finite-measurement characterisations [32–35]. Those results are cited as neighbouring theorem strategies, not as proofs of stronger measure-theoretic or categorical hypotheses.

When the reference  $q$  is uniform on a finite support, relative entropy differs from Shannon entropy only by additive and multiplicative constants [31, 36]. Entropy is therefore not introduced here as a synonym for ignorance. It is the uniform-reference case of recursively decomposable statistical law-content.

Real laws also contain visible constraints. Suppose the retained representation carries observables  $A_i$  with constraints

$$\text{Tr}(\rho A_i) = a_i. \quad (15)$$

The Lagrange multipliers  $g_i$  define a generator

$$K = \sum_i g_i A_i. \quad (16)$$

Thus  $K$  is not primitive in this construction. It is the Legendre-dual encoding of visible statistical constraints.

**Proposition 3** (Free-energy law-form). *For finite states with compatible support and  $M > 0$ , recursively decomposable law-content plus visible constraints gives the variational family*

$$J_K[\rho] = \text{Tr}(\rho K) - M S(\rho), \quad (17)$$

or, classically,

$$\mathcal{F}_M[p] = \langle E \rangle_p + M D_{\text{KL}}(p||q). \quad (18)$$

Minimising (18) with normalisation enforced by  $\alpha$  gives

$$0 = \delta \left[ \sum_x p(x) E(x) + M \sum_x p(x) \log \frac{p(x)}{q(x)} - \alpha \left( \sum_x p(x) - 1 \right) \right]. \quad (19)$$

Solving the Euler equation yields

$$\begin{aligned} p_M(x) &= \frac{1}{Z(M)} q(x) \exp[-E(x)/M], \\ Z(M) &= \sum_x q(x) \exp[-E(x)/M], \end{aligned} \quad (20)$$

with minimum value

$$\mathcal{F}_M^* = -M \log Z(M). \quad (21)$$

This is the free-energy form. The claim is not that thermodynamics is assumed; it is that the form is forced once recursive statistical content and visible constraints are combined. This places the construction beside MaxEnt, MaxCal, large-deviation, and thermodynamic-formalism uses of the same variational grammar [14–18].

Having obtained the invariant form, we can now put familiar effective theory inside it. Coarse-graining no longer changes the kind of law; it changes the generator  $K$  and hence the coordinates used to describe the retained constraints.

## V. RG FLOW AND THE RANK ENSEMBLE

Ordinary RG can now be placed in the construction without confusing two different fixedness claims. The free-energy family is a fixed form. RG fixed points are fixed points of the induced generator flow inside that form [1–6].

Split variables into retained variables  $s$  and hidden variables  $e$ . For a finite hidden fibre with measure  $\mu_s(e)$ ,

$$K_{\text{eff}}(s) = -M \log \sum_e \mu_s(e) \exp[-K(s, e)/M]. \quad (22)$$

In a quantum or open-system notation the analogous expression is

$$K_S^{\text{eff}} = -M \log \text{Tr}_E \exp[-K_{SE}/M] + \text{const}. \quad (23)$$

Equations (22) and (23) are contractions of the generator, not contractions of the law-form itself. Once expanded in a visible operator basis  $K = \sum_a g_a O_a$ , they induce the familiar coupling flow  $g_a \mapsto g'_a$ . In field-theory language, the retained operator basis and truncation boundary are part of the effective description [7, 8].

Figure 4 organizes the finite calculation. It shows where the pieces of an effective description live: law-form invariance at the top, generator flow in the visible projection, and unabsorbed residuals in the complement.

As a concrete finite example, decimate the middle spin in a three-spin Ising block. Let  $s_1, s_2, e \in \{\pm 1\}$ , retain  $s = (s_1, s_2)$ , and take

$$K(s_1, e, s_2) = -J(s_1 e + e s_2). \quad (24)$$

Contracting over  $e$  gives

$$K_{\text{eff}}(s_1, s_2) = -M \log \sum_{e=\pm 1} \exp \left[ \frac{J(s_1 e + e s_2)}{M} \right]. \quad (25)$$

Thus

$$K_{\text{eff}}(s_1, s_2) = -M \log \left[ 2 \cosh \left( \frac{J(s_1 + s_2)}{M} \right) \right]. \quad (26)$$

Projecting this exact contracted generator back to the visible basis  $\{1, s_1 s_2\}$  gives

$$K_{\text{eff}}(s_1, s_2) = A - J' s_1 s_2, \quad J' = \frac{M}{2} \log \cosh \left( \frac{2J}{M} \right), \quad (27)$$

with  $A = -M \log 2 - \frac{M}{2} \log \cosh(2J/M)$ . This is a complete miniature of the claim: eliminating a hidden refinement preserves the free-energy form, but changes the generator coordinate  $J$ . In larger theories the same step produces many couplings, relevant and irrelevant directions, and truncation residuals. The finite block keeps the algebra explicit.

The same calculation becomes an actual recursion for the one-dimensional nearest-neighbour Ising chain. Write  $k = J/M$  and decimate every other spin. For each retained pair  $s_L, s_R$ ,

$$\sum_{e=\pm 1} \exp[k e(s_L + s_R)] = \exp[A(k) + k' s_L s_R], \quad (28)$$

so comparison of equal and opposite retained spins gives

$$e^{2k'} = \cosh(2k), \quad k' = \frac{1}{2} \log \cosh(2k), \quad (29)$$

or equivalently

$$\tanh k' = \tanh^2 k. \quad (30)$$

Thus a repeated coarse-graining step generates a concrete coupling flow. In one dimension the finite-temperature flow runs toward  $k = 0$ , so this is not a nontrivial critical fixed-point calculation. Its role is narrower: it shows the distinction between preserving the free-energy law-form and flowing within the generator coordinates.

**Proposition 4** (RG inside the invariant law-form). *Free energy is an invariant variational law-form under contraction over hidden refinements. Ordinary RG studies the induced flow of  $K$ , or of its couplings, inside this family.*

Why KL? — Ratios  $r = p/q$  compose as products; information must add. The log is forced:  $\phi(ab) = \phi(a) + \phi(b)$  iff  $\phi = \text{Clog}$  (Cauchy). Rank ensemble  $P_I(n) \propto n^{-I}$  makes recursive self-similarity visible:  $\log r(n) = (I_q - I_p) \log n$  repeats identically at every coarse-graining level.

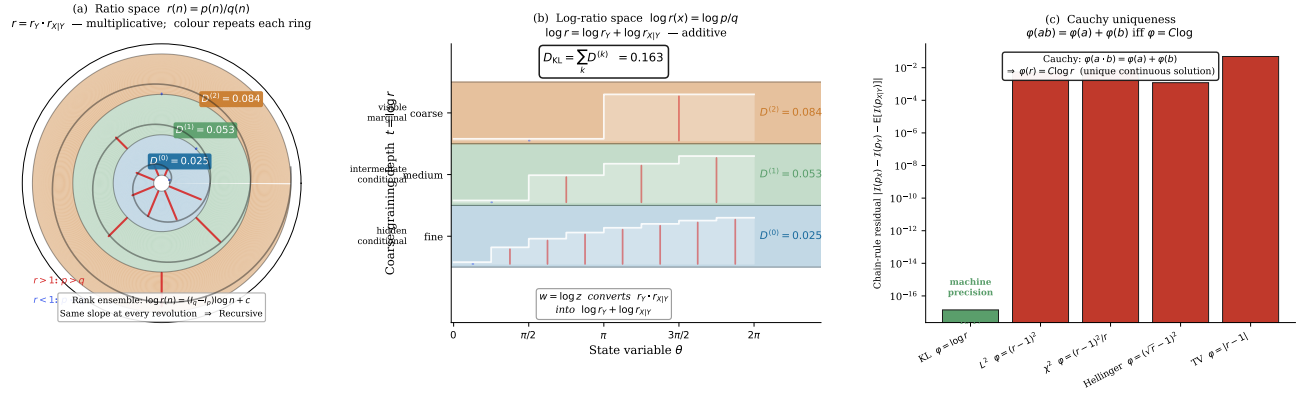


FIG. 3. **The log kernel is forced by recursive self-similarity.** (a) Probability ratios  $r(n) = p(n)/q(n)$  shown as radial bars in a log-spiral arrangement. The rank-ensemble choice  $P_I(n) \propto n^{-I}$  gives  $\log r(n) = (I_q - I_p) \log n + \text{const}$ : the colour sweep is identical at every revolution of the spiral because the log-ratio is linear in  $\log n$ . This is recursive self-similarity by inspection, not by assertion. (b) The conformal map  $w = \log z$  unrolls the spiral into a periodic strip. Each horizontal band contributes one term  $D^{(k)} = \mathbb{E}_p[\log r]$  in the chain rule; the total is their vertical sum. Self-similarity in the spiral  $\Leftrightarrow$  periodicity in the strip  $\Leftrightarrow$  recursive decomposability of law-content. (c) Any admissible law-content  $\mathcal{I}(p||q) = \sum_x p_x \phi(p_x/q_x)$  must satisfy the chain rule, which requires  $\phi(r_Y \cdot r_{X|Y}) = \phi(r_Y) + \phi(r_{X|Y})$  for all splits. Cauchy's functional equation on the positive reals forces  $\phi(r) = C \log r$ . Panel (c) plots the chain-rule residual for five kernels on the rank-ensemble finite diagnostic example: only KL reaches machine precision. Every other kernel fails by an  $O(1)$  amount.

This interpretation also changes the status of effective potentials. A term that appears energetic at the retained scale may be the free-energy remainder of unresolved conditional structure:

$$V_{\text{eff}}(s) = -M \log \int_{\pi^{-1}(s)} e^{-K_{\text{fine}}(s,e)/M} d\mu_s(e). \quad (31)$$

The numerical fibre-contraction example gives  $K_{\text{eff}} = [0.48667, 0.88161]$  for two retained states. The example is minimal: it demonstrates the compression mechanism, not a physical beta function.

The rank ensemble is a useful universal special case. Let alternatives be ranked by  $n$ , and assign canonical complexity

$$K(n) = \log n. \quad (32)$$

Maximising entropy at fixed  $\langle K \rangle$ , or equivalently minimising the corresponding free energy, gives

$$P_I(n) = \frac{n^{-I}}{Z(I)}, \quad Z(I) = \sum_{n=1}^{\infty} n^{-I}. \quad (33)$$

For the infinite rank ensemble,

$$Z(I) = \zeta(I), \quad I > 1. \quad (34)$$

Defining  $\phi(I) = \log Z(I)$ , one obtains

$$\langle K \rangle_I = -\phi'(I), \quad H(I) = \phi(I) - I\phi'(I). \quad (35)$$

Here  $I$  is inverse-temperature-like, with  $M = 1/I$  where that convention is useful. The finite numerical scan

records the expected warning near  $I = 1$ : at  $I = 1.1$ , tail-corrected estimates give  $\langle K \rangle_I \approx 9.4410$  and  $H(I) \approx 12.7445$ .

Up to this point the theory has been about stable law-forms. The next step is different. We ask what happens when the law-form is not written from outside the world, but implemented by a finite coarse register inside it.

## VI. FINITE SELF-DESCRIPTION AND RECONSTRUCTION FREEDOM

The first half of the argument concerned law-forms under changes of resolution. The second half asks what happens when the theory itself is a finite representation produced inside the world it describes. This is the reflexive regime sketched in Fig. 1: the register carrying the model is part of the target being modelled. Let  $X_\mu$  be a fine state space and  $X_M$  a macro-description. A finite description is a lossy map

$$C : X_\mu \rightarrow X_M. \quad (36)$$

For a macrostate  $m \in X_M$ , the fibre  $C^{-1}(m)$  contains fine states that the macro-description cannot distinguish.

To compare or evolve macrostates by using fine dynamics, one often needs a reconstruction map

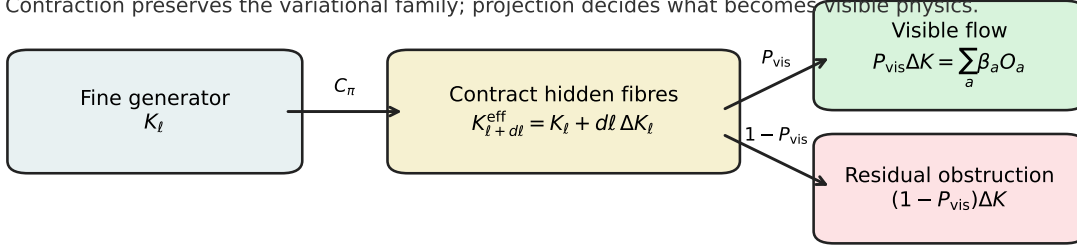
$$R : X_M \rightarrow X_\mu, \quad CR = I_M. \quad (37)$$

The important fact is that  $R$  is not unique. If  $N$  maps into the kernel of  $C$ , then

$$R' = R + N, \quad CN = 0 \quad (38)$$

### Obstruction flow inside an invariant law-form

Contraction preserves the variational family; projection decides what becomes visible physics.



### Scale composition

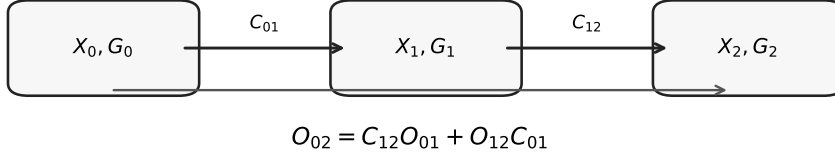


FIG. 4. Obstruction flow inside the invariant free-energy law-form. Contraction over hidden refinements preserves the variational family but changes the generator by  $\Delta K$ . Projection onto the visible operator basis gives beta-function coordinates, while the orthogonal remainder is the residual obstruction. The lower line records the scale-composition identity that makes such residuals law-like rather than arbitrary.

is another reconstruction. The hidden object is therefore not a catalogue of primitive values. It is the fibre freedom erased by the coarse description.

**Proposition 5** (Reconstruction freedom). *Coarse-graining creates fibres; reconstruction chooses sections; the non-uniqueness  $R' = R + N$ ,  $CN = 0$ , is gauge-like freedom induced by erasure.*

The qualifier “gauge-like” is a technical restriction. Equation (38) defines an affine family of right inverses. It becomes a principal bundle only after one specifies a structure group acting on reconstruction choices [37]. Until then, the finite result is reconstruction non-uniqueness, not a full gauge theory.

Let  $G_\mu$  be the fine generator and  $G_M$  the candidate macro-generator. Exact descent requires the square to commute:

$$CG_\mu = G_MC. \quad (39)$$

When this holds, evolve-then-describe equals describe-then-evolve. The macro variables form an autonomous quotient.

The obstruction to exact finite self-description is

$$\mathcal{O} = CG_\mu - G_MC. \quad (40)$$

This is not merely numerical error. It records the part of fine evolution that cannot be represented by the proposed macro-generator.

Different regimes correspond to different treatments of  $\mathcal{O}$ . If  $\mathcal{O} = 0$ , the quotient is exact and macro-variables are autonomous. If  $\mathcal{O}$  is small, the theory is effectively classical. If it can be absorbed into  $G_M$ , it appears as renormalisation (beta-function flow). If it is entropic, it becomes statistical mechanics (heat bath variables returning as thermodynamic forces). If it is non-flat under reconstruction, it becomes gauge or holonomy (Berry phase in the compact case). If it is history-dependent, it becomes a memory kernel or influence functional (Nakajima–Zwanzig residue, see Appendix A). Table II collects all these regimes in statistical-mechanics language.

## VII. COMPOSITION, GAUGE, AND HOLONOMY

Obstructions must themselves compose coherently if finite effective physics is to exist. Consider a tower

$$X_0 \xrightarrow{C_{01}} X_1 \xrightarrow{C_{12}} X_2 \quad (41)$$

with generators  $G_0, G_1, G_2$ . Define

$$\mathcal{O}_{01} = C_{01}G_0 - G_1C_{01}, \quad \mathcal{O}_{12} = C_{12}G_1 - G_2C_{12}. \quad (42)$$

The two-step obstruction is

$$\mathcal{O}_{02} = C_{12}C_{01}G_0 - G_2C_{12}C_{01}. \quad (43)$$



RG as conformal map:  $k' = \frac{1}{2} \log \cosh(2k)$  is  $x \mapsto x^2$  in  $\tanh$  coordinates — log self-similarity throughout

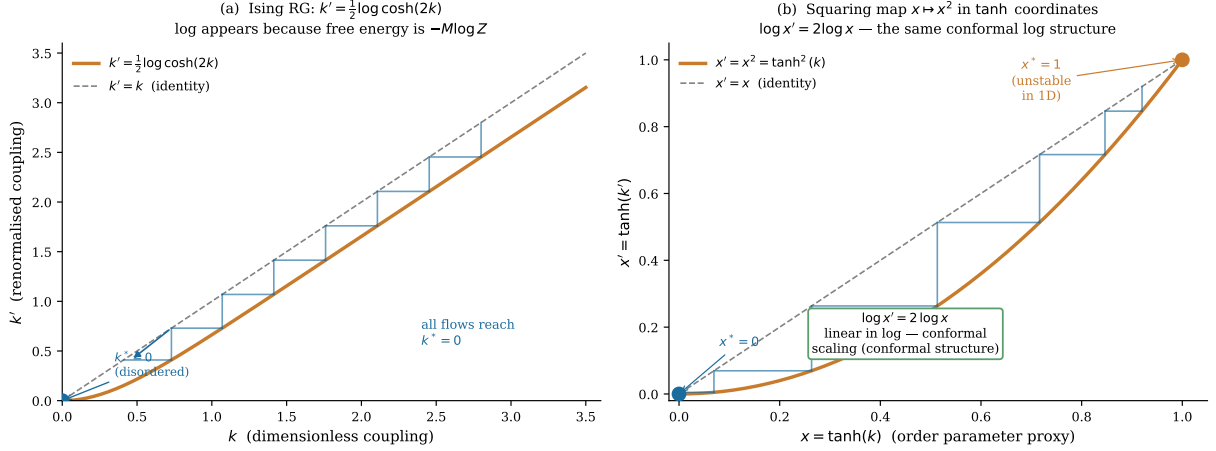


FIG. 5. **RG as conformal map: log self-similarity in coupling space.** (a) Cobweb diagram for the 1D Ising recursion  $k' = \frac{1}{2} \log \cosh(2k)$ . The log appears here for the same reason it appears in the KL kernel: it is the generator flow inside the invariant free-energy family  $K_{\text{eff}}(s) = -M \log \sum_e \mu_s(e) \exp[-K(s, e)/M]$ . All finite-temperature trajectories flow to  $k^* = 0$ . (b) In  $\tanh$  coordinates  $x = \tanh k$ , the recursion reduces to the squaring map  $x \mapsto x^2$ . In log-coordinates:  $\log x' = 2 \log x$ , a linear map—the same conformal scaling structure that appears in the conformal log-spiral of Figure ???. The squaring map is to RG what the log-spiral is to the chain rule: both encode self-similar contraction toward a fixed point.

Adding and subtracting  $C_{12}G_1C_{01}$  gives

$$\mathcal{O}_{02} = C_{12}\mathcal{O}_{01} + \mathcal{O}_{12}C_{01}. \quad (44)$$

The finite matrix demonstration verifies this identity to residual  $3.93 \times 10^{-17}$ .

**Proposition 6** (Obstruction composition). *A law is not only something invariant under coarse-graining. It is something whose failure to descend composes coherently across scale.*

This is the formal bridge back to RG. If obstructions can be reabsorbed into visible couplings, they generate coupling flow. If they cannot, they appear as memory, nonlocal terms, gauge curvature, or contextuality.

Erasure also creates redundancy. If  $C(x) = C(y)$ , then  $x$  and  $y$  are equivalent at the macro level. Transformations that move states within fibres are symmetries of the coarse description, even if they are not primitive symmetries of the underlying world. The symmetry is induced by what the coarse description cannot distinguish.

Let  $\mathcal{R}$  now denote a space of representations, with coarse maps  $C_r$  and reconstructions  $R_r$ . Because  $R_r$  is non-unique, comparing reconstructions at neighbouring  $r$  requires a connection

$$D_i = \partial_i + A_i. \quad (45)$$

Its curvature is

$$\Omega_{ij} = [D_i, D_j] = \partial_i A_j - \partial_j A_i + [A_i, A_j]. \quad (46)$$

Around a closed loop  $\gamma$ , the holonomy is

$$U_\gamma = \mathcal{P} \exp \oint_\gamma A. \quad (47)$$

If a compact one-dimensional reversible sector is isolated, the holonomy may reduce to  $U_\gamma = e^{i\theta_\gamma}$ . Phase is therefore a limiting representation of reconstruction holonomy, not an assumption placed at the beginning. This is the same geometric role played by Berry holonomy and its non-Abelian extension in quantum systems [19–21].

The same obstruction logic also has an algebraic face. If two finite descriptions erase different distinctions, their projection-reconstruction operations need not commute. This gives the minimal bridge from reconstruction freedom to quantum-like kinematics.

## VIII. INCOMPATIBLE ERASURES AND QUANTUM-LIKE KINEMATICS

Noncommutation can arise before one invokes microscopic quantum structure. Let the microspace be  $\Omega = \{1, 2, 3\}$ , and work on distributions in  $\mathbb{R}^3$ . Define two averaging projections. Description  $A$  identifies  $\{1, 2\}$  and leaves  $\{3\}$  alone:

$$\Pi_A = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (48)$$

Descent obstruction  $\mathcal{O} = CG - GC$  — exact closure fails; gap accumulates across scales as a 1-cocycle

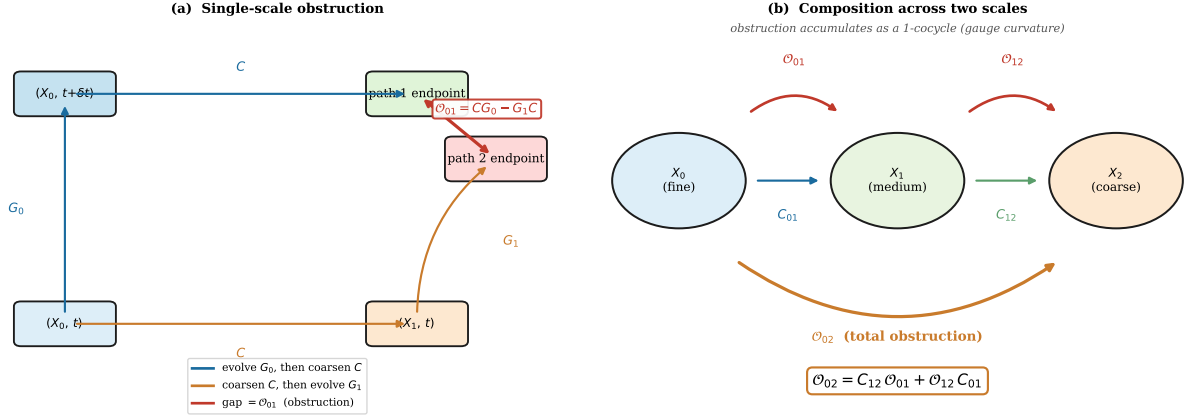


FIG. 6. **Descent obstruction as a non-closing square; accumulation as a 1-cocycle.** (a) Two paths from  $(X_0, t)$  to the coarse level at  $t + \delta t$ : evolve by  $G_0$  then coarsen by  $C$  (blue), versus coarsen first then evolve by  $G_1$  (orange). The gap between the endpoints is the obstruction  $\mathcal{O}_{01} = CG_0 - G_1C$ . Exact descent would require the square to close; it does not. (b) Three-scale composition identity  $\mathcal{O}_{02} = C_{12}\mathcal{O}_{01} + \mathcal{O}_{12}C_{01}$ . The total obstruction across two steps is not simply the sum of the two individual obstructions; it picks up a cross term from the coarse map  $C_{12}$  acting on  $\mathcal{O}_{01}$ . This is a 1-cocycle condition: obstruction accumulates like gauge curvature, not like a scalar error. The numerical verification gives residual  $3.93 \times 10^{-17}$ .

Description  $B$  leaves  $\{1\}$  alone and identifies  $\{2, 3\}$ :

$$\Pi_B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix}. \quad (49)$$

Each operation is classical averaging. Nevertheless

$$[\Pi_A, \Pi_B] = \begin{pmatrix} 0 & -1/4 & 1/4 \\ 1/4 & 0 & -1/4 \\ -1/4 & 1/4 & 0 \end{pmatrix} \neq 0. \quad (50)$$

The Frobenius norm of this commutator is 0.61237 in the finite diagnostic. The fluctuation subspace

$$W = \{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\} \quad (51)$$

reduces the two operations to rank-one projections onto distinct lines, with angle  $\theta = \pi/3$  in the chosen basis.

**Proposition 7** (Incompatible erasures). *Classical lossy descriptions can produce noncommuting projection-reconstruction operations when their erased distinctions are incompatible.*

The consequence is algebraic. If macro-events are represented by noncommuting projection-reconstruction operations, their event algebra is non-Boolean, echoing the quantum-logic boundary identified by Birkhoff and von Neumann [25]. A single joint distribution over primitive properties is then not the natural calculus. Instead, one is led to states as positive normalised functionals over a

noncommutative algebra. In finite-dimensional Hilbert-space language this resembles density-operator calculus:

$$\rho \geq 0, \quad \text{Tr } \rho = 1, \quad p(A) = \text{Tr}(\rho \Pi_A). \quad (52)$$

Sequential update takes the projection form

$$\rho \mapsto \frac{\Pi_A \rho \Pi_A}{\text{Tr}(\rho \Pi_A)}. \quad (53)$$

This paper does not derive the Born rule. Gleason-type results show what additional assumptions can do once one already has the appropriate Hilbert-space projector structure [38]. The present claim is weaker: incompatible finite descriptions provide a route to noncommutative probability as a natural language.

Thus the finite model identifies a kinematic boundary, not a complete quantum reconstruction. The geometric question is what structure keeps track of the cost of moving between descriptions and the curvature of their reconstruction freedom.

## IX. GEOMETRY, MEASUREMENT, AND NO-GO THEOREMS

The preceding sections define a free-energy family over representation space. At each  $r \in \mathcal{R}$ , one may have a coarse map  $C_r$ , reconstruction choices  $R_r$ , a reference measure  $q_r$ , and a generator  $K_r$ . The local functional is

$$\mathcal{F}_{M,r}[p] = \langle E_r \rangle_p + MD_{\text{KL}}(p||q_r), \quad (54)$$

or, in density-operator notation,

$$J_{K_r}[\rho_r] = \text{Tr}(\rho_r K_r) - MS(\rho_r). \quad (55)$$

The symmetric geometry is the local cost of moving through this family. In a coordinate patch one expects a metric from second variations,

$$g_{ij} \sim \partial_i \partial_j \mathcal{F}. \quad (56)$$

Information geometry makes this precise for statistical models through Fisher geometry and its relatives [22]. The antisymmetric sector comes from the reconstruction connection: it is the curvature  $\Omega_{ij}$  in Eq. (46). The schematic combined object is

$$Q_{ij} = g_{ij} + i\Omega_{ij}. \quad (57)$$

The quantum geometric tensor is evidence that this split is physically familiar [23, 24]. Appendix A sharpens the status of (57): the symmetric sector follows from the local KL expansion, while the antisymmetric sector requires an additional reconstruction connection.

Measurement can be described without making an observer fundamental. The observer or apparatus is the subsystem whose internal projection-reconstruction machinery has been compressed into a measurement context. A record is stable when many internal distinctions have become irrelevant to the macro-description. Decoherence and einselection describe how such stable sectors arise in ordinary quantum models [39, 40].

Bell and Kochen-Specker no-go theorems enter here as boundary conditions on global reconstruction. A local hidden-variable model assumes a hidden state  $\lambda$  and outcome functions for all measurement settings, such as

$$A_a(\lambda), \quad A_{a'}(\lambda), \quad B_b(\lambda), \quad B_{b'}(\lambda). \quad (58)$$

This is a flat global reconstruction: one section assigning primitive values to all contexts. Bell's theorem rules out the local version compatible with quantum correlations [26]. Kochen-Specker rules out the single-system non-contextual version [27, 29]. Sheaf-theoretic contextuality makes the gluing obstruction explicit [28].

The present framework is not a local hidden-variable theory. Its hidden object is not a value catalogue. It is non-unique reconstruction freedom created by finite coarse-graining. The no-go theorems are therefore read as obstruction theorems for flattening that freedom globally, not as completed derivations of quantum probability.

## X. DISCUSSION AND OUTLOOK

The central claim can be stated compactly. A finite theory can exist only when the generator it describes admits approximately closed representations:

$$\begin{aligned} &\text{finite self-description possible} \\ &\quad \Rightarrow \\ &\text{approximately closed representations.} \end{aligned} \quad (59)$$

If the theory is imperfect but stable, then its obstructions to closure must themselves be law-governed:

$$\begin{aligned} &\text{stable imperfect self-description} \\ &\quad \Rightarrow \\ &\text{representable obstructions.} \end{aligned} \quad (60)$$

Physics, in this reading, studies representations whose losses remain structured enough to be represented again.

The v2 manuscript therefore has a sharper claim than the earlier notes. It derives the finite relative-entropy/free-energy law-form, shows that ordinary RG is generator flow within that form, proves the affine structure of reconstruction freedom, proves the composition law for descent obstructions, and displays exact finite residues that become memory, induced couplings, or non-commuting descriptions. The compact diagnostics in Fig. 7 record the finite identities on which this claim rests.

The remaining assumptions are now localized. The KL result is finite and positive-support: extending it to measurable spaces or categorical settings requires domain and regularity control. Free-energy contraction closure is proved for finite fibres; field-theoretic versions require a measure, renormalisation scheme, and operator-basis control. Reconstruction freedom is an affine bundle until a structure group and connection are supplied. The route from incompatible finite descriptions to Hilbert-space kinematics requires additional reconstruction axioms, and the Born rule is not derived in this paper.

That boundary is a strength of the present version. It leaves the paper with a definite theorem, a definite obstruction calculus, and a clear list of extra assumptions needed for any more ambitious quantum reconstruction.

TABLE III. Limiting regimes of the cost and obstruction geometry. The table summarizes how a single finite calculus connects several standard effective descriptions; it is not a classification theorem.

| Regime                    | Condition                         | Familiar limit               |
|---------------------------|-----------------------------------|------------------------------|
| Exact quotient            | $\mathcal{O} = 0$                 | autonomous classical closure |
| Small obstruction         | $\ \mathcal{O}\ $ negligible      | effective classical theory   |
| Absorbable obstruction    | visible-basis reexpansion         | RG flow and universality     |
| Entropic residue          | hidden alternatives dominate      | statistical mechanics        |
| Scalar KL cost            | free-energy contraction           | variational thermodynamics   |
| Metric sector             | second variation of $\mathcal{F}$ | information geometry         |
| Non-flat reconstruction   | $\Omega_{ij} \neq 0$              | gauge and holonomy           |
| Compact reversible fibre  | $U_\gamma = e^{i\theta}$          | phase-like structure         |
| Incompatible erasures     | $[\Pi_A, \Pi_B] \neq 0$           | contextual event algebra     |
| History-dependent residue | no Markovian retained closure     | memory kernels               |
| Redundant records         | stable pointer sectors            | classical measurement        |

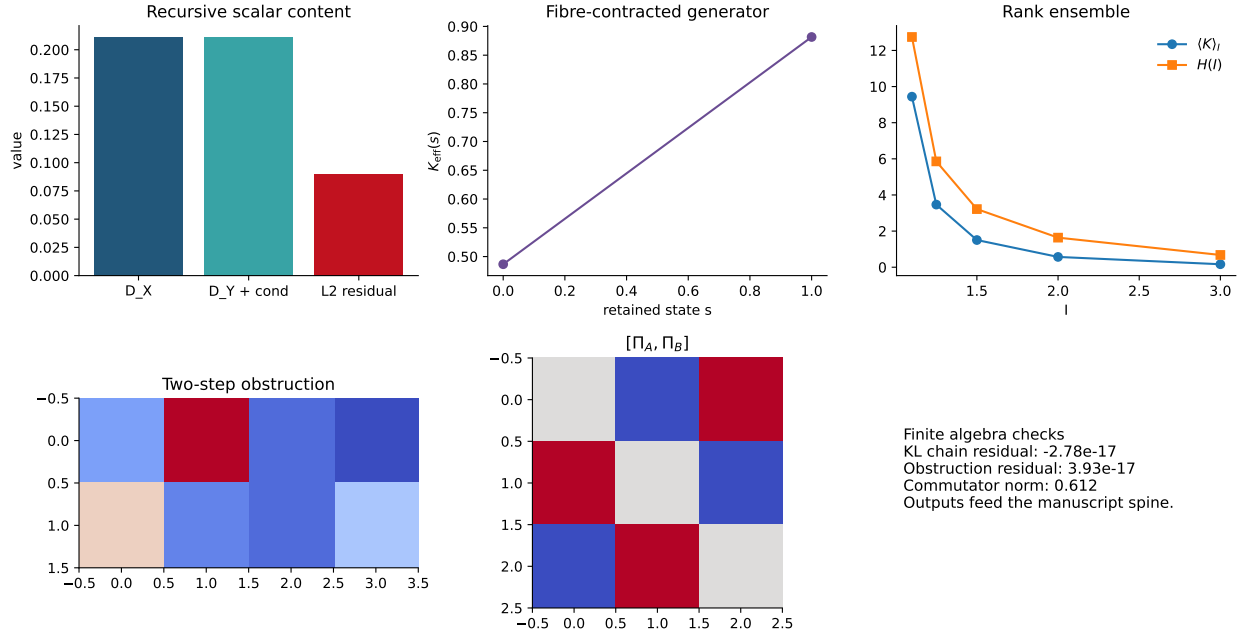


FIG. 7. Finite diagnostics for the algebraic spine. The panels summarize the KL chain-rule check, finite fibre contraction, rank-ensemble behaviour, obstruction-composition identity, and the three-state noncommuting-erasure example. These are checks of the finite identities used in the paper, not phenomenological simulations.

## Appendix A: Finite Technical Resolutions

This section separates results that can already be proved in the finite model from claims that still require stronger structure. The distinction matters. The finite algebra should be proved explicitly, while quantum reconstruction claims should enter only after the needed extra assumptions have been named.

### 1. Local Form From Relabelling And Neutral Refinement

**Proposition 8** (Finite local form from locality). *Let  $X$  be finite and let  $\mathcal{I}_X(p||q)$  be a continuous scalar law-content for positive probability vectors. Assume:*

1. Relabelling invariance: *bijections of  $X$  do not change  $\mathcal{I}_X$ .*

2. Atomic locality: *disjoint alternatives contribute through one-atom terms, where an atom is an elementary outcome (a single sample point of the probability space  $X$ , the analogue of a microstate in statistical mechanics), so before relabelling one may write  $\mathcal{I}_X(p||q) = \sum_{i \in X} \ell_i(p_i, q_i)$ .*

3. Neutral-refinement invariance: *if an atom of masses  $(a, b)$  is split into subatoms  $(\alpha_j a, \alpha_j b)$ , with  $\alpha_j > 0$  and  $\sum_j \alpha_j = 1$ , the content assigned to that atom is unchanged.*

Then

$$\mathcal{I}_X(p||q) = \sum_{i \in X} p_i \varphi\left(\frac{p_i}{q_i}\right) \quad (\text{A1})$$

for a continuous function  $\varphi$  on  $\mathbb{R}_+$ .

*Proof.* Atomic locality gives a sum of one-atom contributions. Relabelling any two atoms exchanges their names but not the value of the scalar content, so the same local density  $\ell$  may be used for every atom:

$$\mathcal{I}_X(p||q) = \sum_i \ell(p_i, q_i).$$

Neutral refinement now imposes the missing scale law. For any fixed likelihood ratio  $r = a/b$ , define

$$g_r(t) = \ell(rt, t), \quad t > 0.$$

Splitting a cell of  $q$ -mass  $t_1 + t_2$  into two neutral subcells with  $q$ -masses  $t_1$  and  $t_2$ , and the same ratio  $r$ , gives

$$g_r(t_1 + t_2) = g_r(t_1) + g_r(t_2).$$

Continuity excludes the pathological Cauchy solutions, hence

$$g_r(t) = t g_r(1).$$

Therefore

$$\ell(a, b) = b g_{a/b}(1) = a \varphi(a/b), \quad \varphi(r) = \frac{g_r(1)}{r}.$$

Substituting  $(a, b) = (p_i, q_i)$  for each atom gives (A1).  $\square$

This is the finite analogue of the log-rank move in the Cost of Complexity argument. Relabelling removes arbitrary names; a locality requirement then decides what survives under admissible refinement. In the rank case, composition turns multiplicative rank into additive logarithmic cost. Here, neutral refinement turns absolute cell masses into a probability-weighted function of the local likelihood ratio.

### 2. KL Uniqueness in the Finite Local Class

**Proposition 9** (Finite local-separable KL uniqueness). *Let  $X$  be finite, let  $p_i > 0$ ,  $q_i > 0$ , and suppose scalar law-content has the local ratio-separable form*

$$\mathcal{I}(p||q) = \sum_i p_i \varphi\left(\frac{p_i}{q_i}\right), \quad (\text{A2})$$

where  $\varphi$  is continuous. If  $\mathcal{I}$  obeys the refinement chain rule (12) for all finite refinements, then

$$\mathcal{I}(p||q) = C \sum_i p_i \log \frac{p_i}{q_i} \quad (\text{A3})$$

for some constant  $C$ . Positivity fixes  $C \geq 0$ , and non-triviality fixes  $C > 0$ .

*Proof.* Refine a coarse state  $i$  into conditional alternatives  $j$ . Write

$$p_{ij} = p_i r_{j|i}, \quad q_{ij} = q_i s_{j|i},$$

with all probabilities positive. Set  $a_i = p_i/q_i$  and  $b_{j|i} = r_{j|i}/s_{j|i}$ . The direct fine content is

$$\sum_{ij} p_{ij} r_{j|i} \varphi(a_i b_{j|i}).$$

The chain rule requires this to equal

$$\sum_i p_i \varphi(a_i) + \sum_i p_i \sum_j r_{j|i} \varphi(b_{j|i})$$

for arbitrary positive refinements. Hence

$$\varphi(ab) = \varphi(a) + \varphi(b)$$

on positive  $a, b$ . Continuity on  $\mathbb{R}_+$  gives  $\varphi(a) = C \log a$ . Substitution gives (A3). The usual log-sum inequality then gives positivity for  $C \geq 0$ , with equality only when  $p = q$  for  $C > 0$ .  $\square$

Together with Proposition 8, this proves the finite positive-support claim from relabelling, atomwise locality, neutral refinement, continuity, and the staged chain rule. The resulting local form is close to the classical

information-type divergence setting [30], and it sits beside independent relative-entropy characterisations based on prior-to-posterior updating and functorial finite measurement maps [33, 35]. Those references are not being used as a black-box proof of the present theorem. Rather, they fix the status of the result: the finite proof is complete, while a stronger paper must extend it beyond finite positive supports or reformulate the axioms to match one of the known characterisation theorems exactly.

### 3. Contraction Closure and Visible Generator Flow

**Proposition 10** (Finite free-energy contraction closure). *Let  $\pi : X \rightarrow Y$  be a finite coarse map and decompose the reference measure as  $q_X(x) = q_Y(y)q_{X|Y}(x|y)$  for  $x \in X_y$ . For*

$$\mathcal{F}_X[p_X] = \sum_x p_X(x)E_X(x) + MD_{\text{KL}}(p_X||q_X), \quad (\text{A4})$$

with  $M > 0$ ,

the fibre contraction at fixed  $p_Y = \pi_* p_X$  is again a free-energy functional,

$$\inf_{\pi_* p_X = p_Y} \mathcal{F}_X[p_X] = \sum_y p_Y(y)E_{\text{eff}}(y) + MD_{\text{KL}}(p_Y||q_Y), \quad (\text{A5})$$

where

$$E_{\text{eff}}(y) = -M \log \sum_{x \in X_y} q_{X|Y}(x|y)e^{-E_X(x)/M}. \quad (\text{A6})$$

*Proof.* Use the KL chain rule:

$$D_{\text{KL}}(p_X||q_X) = D_{\text{KL}}(p_Y||q_Y) + \sum_y p_Y(y)D_{\text{KL}}(p_{X|Y}(\cdot|y)||q_{X|Y}(\cdot|y)).$$

At fixed  $p_Y$ , the conditional minimisations decouple over fibres. For each  $y$ , minimising

$$\sum_{x \in X_y} p(x|y)E_X(x) + MD_{\text{KL}}(p(\cdot|y)||q_{X|Y}(\cdot|y))$$

gives the Gibbs conditional

$$p^*(x|y) = \frac{q_{X|Y}(x|y)e^{-E_X(x)/M}}{\sum_{x' \in X_y} q_{X|Y}(x'|y)e^{-E_X(x')/M}},$$

and minimum value (A6). Substituting these conditional minima gives (A5).  $\square$

Thus the finite free-energy family is closed under contraction. Ordinary RG is obtained by expanding  $E_{\text{eff}}$  or  $K_{\text{eff}}$  in the retained operator basis and tracking the induced coefficients.

Write the retained generator as  $K_\ell = \sum_a g_a(\ell)O_a$ . For an infinitesimal change of scale,

$$K_{\ell+d\ell}^{\text{eff}} - K_\ell = d\ell \Delta K_\ell + o(d\ell). \quad (\text{A7})$$

If  $P_{\text{vis}}$  denotes projection onto the chosen visible operator basis, then

$$P_{\text{vis}}\Delta K_\ell = \sum_a \beta_a(g)O_a, \quad (\text{A8})$$

while  $(1 - P_{\text{vis}})\Delta K_\ell$  is the truncation or residual obstruction. This is the precise sense in which beta functions are the visible-basis coordinates of an absorbable obstruction.

### 4. Support and Reconstruction Structure

The support convention is also now fixed. In finite spaces,  $D_{\text{KL}}(p||q)$  is finite only when  $q_i = 0$  implies  $p_i = 0$ . In the measure-theoretic version this becomes absolute continuity  $p \ll q$ , with

$$D_{\text{KL}}(p||q) = \int \log\left(\frac{dp}{dq}\right) dp.$$

If  $p \not\ll q$ , the law-content is  $+\infty$ . This convention is not a technical nuisance; it says that a proposed reference reconstruction must give nonzero weight to every refinement the law allows.

**Proposition 11** (Affine reconstruction bundle). *For a fixed linear coarse map  $C : X_\mu \rightarrow X_M$ , the set of right inverses*

$$\text{Sec}(C) = \{R : X_M \rightarrow X_\mu \mid CR = I_M\}$$

*is either empty or an affine space modelled on*

$$\text{Vert}(C) = \{N : X_M \rightarrow X_\mu \mid CN = 0\}.$$

*Proof.* If  $R, R' \in \text{Sec}(C)$ , then  $C(R' - R) = 0$ , so  $R' - R \in \text{Vert}(C)$ . Conversely, if  $R \in \text{Sec}(C)$  and  $N \in \text{Vert}(C)$ , then  $C(R + N) = I_M$ . Thus the sections form an affine translate of the vertical space.  $\square$

Principal-bundle language requires more. One must specify a group

$$G_C = \{U : X_\mu \rightarrow X_\mu \mid CU = C\}$$

or an appropriate subgroup acting freely and transitively on admissible sections. Without that group, the honest object is an affine bundle of reconstruction choices, not yet a principal gauge bundle.

### 5. Composed Obstructions and Memory Residues

The obstruction-composition law also extends cleanly from matrices to Markov kernels, because kernels act linearly on measures. If  $C_{ij}$  are stochastic kernels and  $G_i$  are linear generators on signed measures, then  $\mathcal{O}_{ij} = C_{ij}G_i - G_jC_{ij}$  and the same add-subtract proof gives

$$\mathcal{O}_{02} = C_{12}\mathcal{O}_{01} + \mathcal{O}_{12}C_{01}.$$

For nonlinear deterministic coarse maps, the corresponding infinitesimal statement uses tangent maps:

$$\mathcal{O}_{02}(x) = TC_{12} \mathcal{O}_{01}(x) + \mathcal{O}_{12}(C_{01}x),$$

whenever the maps are differentiable and the generators are vector fields.

Memory kernels are a concrete obstruction residue. Let  $L$  be a microscopic Liouville generator and  $P$  a projection onto retained observables, with  $Q = 1 - P$ . In the Nakajima-Zwanzig/Mori-Zwanzig projection-operator setting [9–11, 13, 41], the retained evolution obeys an exact identity of the form

$$\begin{aligned} \frac{d}{dt}Px(t) &= PLPx(t) \\ &+ \int_0^t PL e^{(t-s)QL} QLPx(s) ds \\ &+ PL e^{tQL} Qx(0). \end{aligned} \quad (\text{A9})$$

The first term is the closed retained generator. The integral and final noise term are precisely the price of eliminating  $Q$ -variables. Markovian closure is the special case in which those residues can be neglected or compressed into a finite enlarged state.

The finite-dimensional version of the same identity is transparent block algebra. Write a linear generator in retained and hidden blocks,

$$G_\mu = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad x(t) = \begin{pmatrix} u(t) \\ v(t) \end{pmatrix}, \quad (\text{A10})$$

where  $u$  is retained and  $v$  is eliminated. Solving the hidden equation and substituting into the retained one gives

$$\dot{u}(t) = Au(t) + Be^{tD}v(0) + \int_0^t Be^{(t-s)D}Cu(s) ds. \quad (\text{A11})$$

Thus the noise is  $Be^{tD}v(0)$ , and the memory kernel is  $K(t) = Be^{tD}C$ . Exact autonomous descent occurs only if these hidden-block terms vanish or can be represented by additional retained variables.

A two-mode linear example makes this residue explicit. Let the fine state be  $(u, v)$ , with  $u$  retained and  $v$  hidden, and take

$$\frac{d}{dt} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & a \\ -b & -\gamma \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}, \quad \gamma > 0. \quad (\text{A12})$$

The coarse description keeps only  $u$ . Solving the hidden equation gives

$$v(t) = e^{-\gamma t}v(0) - b \int_0^t e^{-\gamma(t-s)}u(s) ds. \quad (\text{A13})$$

Substitution into  $\dot{u} = av$  gives the exact retained dynamics

$$\dot{u}(t) = ae^{-\gamma t}v(0) - ab \int_0^t e^{-\gamma(t-s)}u(s) ds. \quad (\text{A14})$$

Thus a finite linear system with a perfectly ordinary generator need not descend to a first-order autonomous equation on the retained coordinate. The hidden coordinate returns as a transient noise term and an exponential memory kernel. When  $\gamma$  is large and  $u$  varies slowly, the kernel contracts to an approximate local term,

$$\dot{u}(t) \approx -\frac{ab}{\gamma}u(t), \quad (\text{A15})$$

after the initial transient has decayed. This is the elementary version of an absorbable obstruction: the exact residue is nonlocal in time, while a separated-scale approximation compresses it into an effective generator.

The same example also makes  $\mathcal{O} = CG_\mu - G_M C$  concrete. Take

$$G_\mu = \begin{pmatrix} 0 & a \\ -b & -\gamma \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad (\text{A16})$$

and let  $G_M$  be any scalar generator on the retained coordinate. Then

$$\mathcal{O} = CG_\mu - G_M C = \begin{pmatrix} -G_M & a \end{pmatrix}. \quad (\text{A17})$$

No choice of scalar  $G_M$  makes  $\mathcal{O} = 0$  when  $a \neq 0$ . The second component is the hidden channel through which  $v(0)$  and the memory kernel return to the retained dynamics. The Markovian generator  $G_M \approx -ab/\gamma$  is therefore not exact descent; it is the scale-separated compression of a nonlocal obstruction residue.

## 6. Metric Sector and Curvature Sector

Finally, the geometry can be sharpened. For a smooth statistical model  $p_r(x)$ , the second-order expansion of KL divergence gives

$$D_{\text{KL}}(p_{r+dr} \| p_r) = \frac{1}{2} g_{ij}(r) dr^i dr^j + O(|dr|^3), \quad (\text{A18})$$

with Fisher metric

$$g_{ij}(r) = \mathbb{E}_{p_r} [\partial_i \log p_r \partial_j \log p_r]. \quad (\text{A19})$$

The antisymmetric sector does not come from a scalar KL Hessian. It appears only after reconstruction choices form a bundle with connection  $A_i$ , whose curvature is  $\Omega_{ij}$ . A complex object

$$Q_{ij} = g_{ij} + i\Omega_{ij}$$

therefore requires two inputs: scalar distinguishability from free-energy/KL and non-flat reconstruction transport. In the compact unitary case this has the same metric-plus-curvature structure familiar from the quantum geometric tensor, but here it is a conditional construction, not a derivation of quantum mechanics.

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